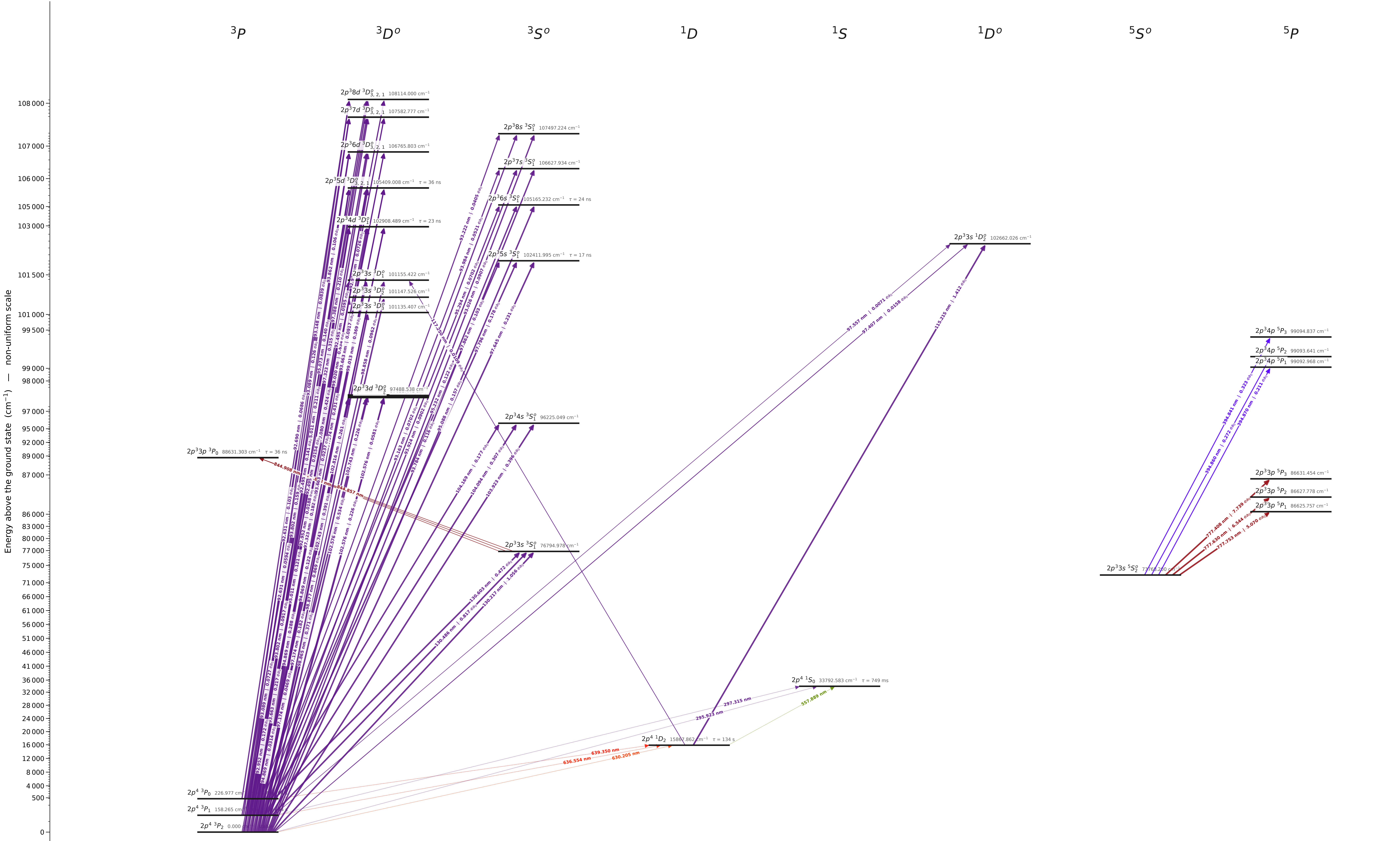
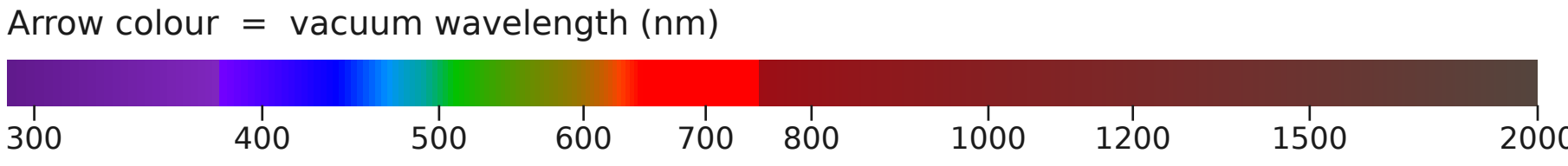


Oxygen

O (O I). Strongest classified transitions below 2 μm (NIST): 43 levels, 79 lines.

O⁺ ionisation limit 109837.02 cm⁻¹ = 13.61805 eV (91.044 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle U||e\mathbf{r}||U'\rangle|$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er U'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2p ⁴ ³ P ₁ – 2p ⁴ ¹ S ₀	297.3154	297.2286	1008.3315	–	7.540e-02	2.126e-07	5.64	ultraviolet auroral line [O I] 297.2 nm
2p ⁴ ¹ D ₂ – 2p ⁴ ¹ S ₀	557.8887	557.7339	537.3696	–	1.260e+00	2.126e-07	94.3	green auroral / airglow line [O I] 557.7 nm
2p ⁴ ³ P ₂ – 2p ⁴ ¹ D ₂	630.2046	630.0304	475.7065	–	5.630e-03	1.189e-09	75.3	red auroral / airglow line [O I] 630.0 nm
2p ⁴ ³ P ₁ – 2p ⁴ ¹ D ₂	636.5536	636.3776	470.9619	–	1.820e-03	1.189e-09	24.3	red auroral / airglow line [O I] 636.4 nm
2p ⁴ ³ P ₀ – 2p ⁴ ¹ D ₂	639.3500	639.1733	468.9019	–	8.600e-07	1.189e-09	0.0115	[O I] 639.2 nm
2p ³ 3s ⁵ S _g ^o – 2p ³ 3p ⁵ P ₃	777.4083	777.1944	385.6307	7.739	3.690e+07	–	–	777.19 nm
2p ³ 3s ⁵ S _g ^o – 2p ³ 3p ⁵ P ₂	777.6305	777.4166	385.5204	6.544	3.690e+07	–	–	777.42 nm
2p ³ 3s ⁵ S _g ^o – 2p ³ 3p ⁵ P ₁	777.7527	777.5388	385.4599	5.070	3.690e+07	–	–	777.54 nm
2p ³ 3s ³ S _g ^o – 2p ³ 3p ³ P ₀	844.8568	844.6247	354.8441	–	–	4.421	–	844.6 nm fluorescence line of the O TALIF scheme: two-pho...
2p ³ 3s ³ S _g ^o – 2p ³ 3p ³ P ₂	844.8680	844.6359	354.8394	–	–	4.421	–	844.6 nm fluorescence line of the O TALIF scheme: two-pho...
2p ³ 3s ³ S _g ^o – 2p ³ 3p ³ P ₁	844.9079	844.6758	354.8226	–	–	4.421	–	844.6 nm fluorescence line of the O TALIF scheme: two-pho...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁷O hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2p ⁴ ³ P ₂	-218.569(4)	-10.44(3)	F=1/2: +1522.68 F=3/2: +1199.52 F=5/2: +658.32 F=7/2: -104.85 F=9/2: -1095.45
2p ⁴ ³ P ₁	4.74(4)	5.20(9)	F=3/2: -12.94 F=5/2: -8.90 F=7/2: +13.14
2p ³ 3s ⁵ S _g ^o	-97.93(10)	–	F=1/2: +685.51 F=3/2: +538.62 F=5/2: +293.79 F=7/2: -48.97 F=9/2: -489.65
2p ³ 3p ⁵ P ₁	-25.83(10)	–	F=3/2: +90.41 F=5/2: +25.83 F=7/2: -64.58
2p ³ 3p ⁵ P ₂	-24.47(11)	0.9(8)	F=1/2: +171.92 F=3/2: +134.81 F=5/2: +73.19 F=7/2: -12.62 F=9/2: -122.12
2p ³ 3p ⁵ P ₃	-18.70(5)	–	F=1/2: +186.88 F=3/2: +158.88 F=5/2: +112.20 F=7/2: +46.81 F=9/2: -37.34 F=11/2: -140.30

Measured A, B constants; sources in the data file.